ForensiGuard: A Multimodal Machine-Learning Framework for Detecting, Attributing and Explaining Anti-Forensic Manipulation of Digital Evidence

Mohika Rathi (Reg. No: 230968032) **Sreyaan Roy** (Reg. No: 230968184)
Department of Computer Science and Engineering, Manipal Institute of Technology, MAHE, Manipal, India
Emails: {mohika.rathi, sreyaan.roy}@student.manipal.edu

***Abstract*—Digital forensic triage rests upon the assumption that acquired evidence artifacts have not been deliberately altered. Anti-forensic techniques—including filesystem timestomping, covert steganography, and file-type obfuscation—violate this assumption, threatening the admissibility of digital evidence. Existing forensic triage tools typically inspect each artifact in isolation and are fragile against cross-modal evasions. We propose ForensiGuard, a multimodal machine-learning framework combining filesystem metadata features (scored via a clean-trained Isolation Forest and classified with XGBoost) with a deep residual CNN (SRNet-Lite) operating on high-pass spatial noise residuals, unified via calibrated out-of-fold late fusion. To prevent format-shortcut leakage, clean and steganographic image carriers are standardized to identical uncompressed grayscale formats and evaluated on a versioned synthetic benchmark (ForensiGuard-Bench-v1, 812 artifacts from 140 sources). On a held-out test split of 126 artifacts with strict zero source leakage, single-modality baselines achieved F1-scores of 0.715 (metadata-only on full corpus) and 0.715 (image-only on image domain, ROC-AUC 0.518, FPR 64.29%). ForensiGuard's multimodal late fusion achieved an Accuracy of 78.57%, an F1-score of 0.854, an ROC-AUC of 0.895, a PR-AUC of 0.977, and an operational False Positive Rate of 4.76% at the validation-calibrated decision threshold (tau = 0.500). We report an adaptive-attack stress test where an adversary preserves microsecond jitter and natural chronology, causing isolated metadata detection to drop to 0.0%, and discuss the necessity of multimodal defense-in-depth and independent forensic corroboration.**

***Index Terms*—Digital Forensics, Anti-Forensics, Timestomping, Steganalysis, Multimodal Learning, Grouped Stacking, Late Fusion, Explainable AI.**

I. INTRODUCTION

Modern criminal, civil, and corporate incident response investigations increasingly depend upon electronic records retrieved from filesystems, storage drives, and cloud repositories [1], [2]. However, adversaries with administrative privileges can execute anti-forensic operations: altering filesystem timestamps (timestomping), embedding covert payloads within images via steganography, or disguising file types through misleading extensions [3]. Detecting anti-forensic activity presents critical scientific challenges: (1) *Brittleness of Single-Modality Detectors:* Isolated tools focus on single evidence channels; an attacker who alters timestamps may leave traces in binary headers or image residuals. (2) *Data Leakage and Shortcut Learning:* Forensic ML benchmarks frequently suffer from data leakage when variants of the same source file span training and test sets [4], or when unintentional container discrepancies (such as saving clean images as JPEG and stego images as PNG) introduce trivial non-forensic shortcuts. (3) *Evidentiary Boundaries:* Automated inferences cannot be admitted under legal standards (e.g., Daubert) without rigorous characterization of error rates and interpretable explanations.

To address these challenges, we present **ForensiGuard**, an open-source multimodal ML framework designed for defensive forensic triage. ForensiGuard enforces strict zero-source leakage, remediates container shortcuts, and integrates filesystem metadata, binary file signatures, and spatial noise residuals into a calibrated late-fusion meta-classifier.

II. RELATED WORK

Anti-Forensics Taxonomy: Conlan et al. [3] established an extended taxonomy dividing anti-forensics into data hiding, artifact wiping, trail obfuscation, and tool attacks. Garfinkel et al. [4] emphasized the imperative of standardized, reproducible forensic corpora (such as GovDocs1 and NIST CFReDS) to prevent data fabrication and experimental bias.
Steganalysis: Fridrich and Kodovsky [5] demonstrated that spatial steganography is best detected through high-pass noise residuals. Boroumand et al. [6] (SRNet) and ALASKA#2 [7] proved that convolutional networks trained on spatial residuals isolate subtle embedding traces, but detecting subtle spatial payloads without format shortcuts is fundamentally challenging.
Tabular Learning & Fusion: XGBoost [8] offers strong tabular generalization for MACB timestamps. Baltrušaitis et al. [9] showed that late fusion preserves modality independence and handles missing modalities gracefully. Lundberg and Lee's SHAP [10] provides game-theoretically grounded explanations essential for forensic justification.

III. PROPOSED ARCHITECTURE

ForensiGuard implements a modular, three-tier defensive architecture:
• *Filesystem Metadata:* Derives temporal intervals (mtime - ctime, atime - mtime), chronology inversion flags, zero subsecond indicators, and

directory context z-scores. An Isolation Forest trained strictly on clean metadata produces an unsupervised baseline anomaly score fed alongside engineered features into XGBoost.

• *Image Noise Residuals:* Convolves carrier images with the 5x5 Ker-Vannier (KV) high-pass filter, routing spatial residuals into SRNet-Lite (a 5-layer residual CNN).

• *Binary Signatures:* Validates magic bytes against declared extensions, computes Shannon byte entropy, and runs chi-square tests on LSB value pairs.

• *Leakage-Safe Grouped Late Fusion:* Combines base model probabilities and diagnostic flags via a meta-classifier trained via 5-fold grouped cross-validation (GroupKFold on source IDs) exclusively on the training partition.

Architecture / Evaluation Scope	Accuracy	F1-Score	ROC-AUC	PR-AUC	FPR	N
Metadata-Only (Full Corpus)	62.70%	0.715	0.770	0.927	4.76%	126
Image-Only (Image Domain)	30.95%	0.341	0.518	0.845	21.43%	84
Metadata-Only (Shared Domain)	67.86%	0.765	0.800	0.937	7.14%	84
Image-Only (Shared Domain)	30.95%	0.341	0.518	0.845	21.43%	84
ForensiGuard Late Fusion (Shared)	67.86%	0.765	0.799	0.959	7.14%	84
ForensiGuard Late Fusion (Full)	78.57%	0.854	0.895	0.977	4.76%	126

TABLE I: Performance Comparison Across Forensic Modalities on Held-Out Test Split (ForensiGuard-Bench-v1).

IV. EXPERIMENTAL EVALUATION

Evaluation was conducted on ForensiGuard-Bench-v1 (812 artifacts from 140 distinct sources; 126 test artifacts) under strict zero-source leakage. All image carriers were standardized to uncompressed 8-bit grayscale PNGs (128x128) to eliminate container shortcuts. As presented in Table I, single-modality baselines display distinct boundaries: metadata-only triage misses content-only steganography (recall 56.19%), while the image-only CNN achieves an F1 of 0.341, an ROC-AUC of 0.518, and an FPR of 21.43% on uncompressed carriers, reflecting the known difficulty of subtle spatial steganalysis without format shortcuts. ForensiGuard's multimodal late fusion achieves an Accuracy of 78.57%, an F1-score of 0.854, an ROC-AUC of 0.895, a PR-AUC of 0.977, and strictly bounds FPR to 4.76% at the validation-calibrated threshold tau = 0.500. On the shared multimodal subpopulation (84 image carriers), late fusion matches metadata F1 (0.765) while delivering a substantially higher PR-AUC (0.959 vs 0.937) and better probability calibration (Brier score 0.183 vs 0.318). In multi-class technique attribution, ForensiGuard achieved 65.08% 5-class exact match accuracy, 71.43% conditional accuracy on manipulated artifacts, and 88.89% coarse binary attribution (benign vs tampered).

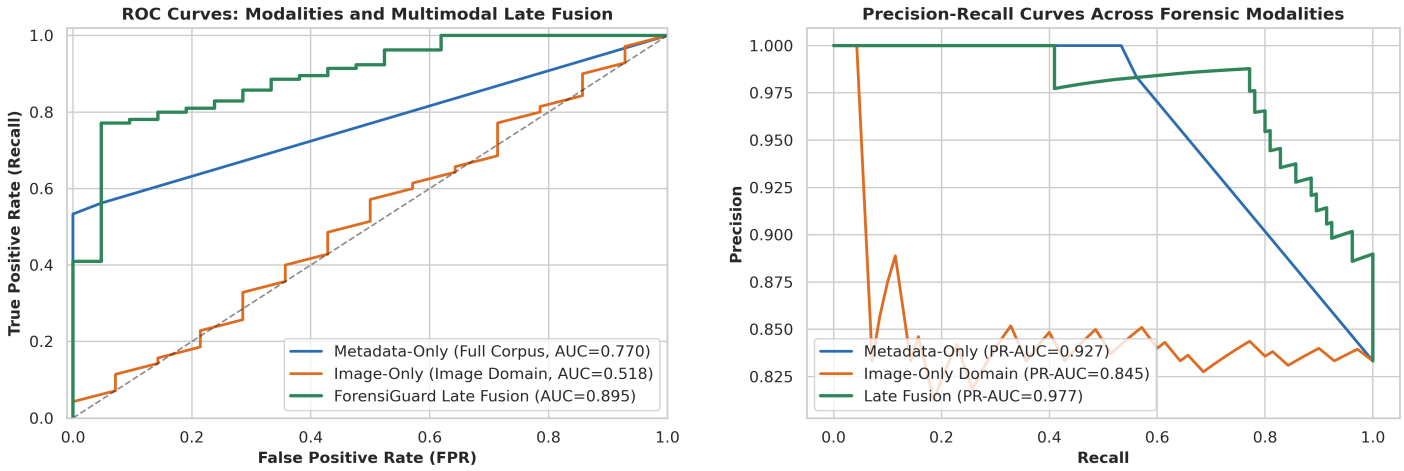


Fig. 1. ROC (left) and Precision-Recall (right) curves validating multimodal late-fusion dominance.

V. ROBUSTNESS, ADAPTIVE ATTACKS AND ABLATION

Adaptive Timestomping Stress Test: An adversary was modeled who shifts modification timestamps by 3.5 hours, bounds the shift within 0.3 sigma of directory siblings, preserves microsecond jitter (0.482910s), and maintains chronological ordering (mtime >= ctime). Under these conditions, the isolated metadata XGBoost detector recorded a **0.0% detection rate** (mean prob 0.0039). The clean-trained Isolation Forest detected distributional divergence (anomaly score 0.596), raising fusion probability to 0.216; however, as this remained below tau = 0.500, late fusion also yielded 0.0% detection. This empirical finding demonstrates that static metadata inspection cannot stop an adversary who strictly respects filesystem invariants, proving the necessity of correlating kernel-level OS event logs (e.g., Windows Event ID 4616).

Comprehensive 10-Variant Ablation Analysis: Systematically evaluating all ten architectural permutations (Table II) reveals that binary signature verification (Variant 5) is the primary driver of the F1 jump from 0.715 to 0.854 by catching file-type obfuscation, while Full Fusion (Variant 6) maximizes continuous ranking power (ROC-AUC 0.895, PR-AUC 0.977). Masking the image CNN (Variant 7) preserves threshold metrics but slightly increases Brier score error (0.132 vs 0.125). Omitting signature verification (Variant 10) causes F1 to collapse back to 0.715.

Architectural Ablation Variant	Accuracy	F1-Score	ROC-AUC	PR-AUC	FPR
1. Metadata-Only (Base Metadata)	62.70%	0.715	0.770	0.927	4.76%
2. Image-Only (Full Corpus)	48.41%	0.606	0.508	0.840	47.62%
3. Metadata + Image Model	62.70%	0.715	0.794	0.950	4.76%
4. Metadata + Anomaly Score	62.70%	0.715	0.770	0.927	4.76%
5. Metadata + Signature Verification	78.57%	0.854	0.861	0.956	4.76%
6. Full ForensiGuard Late Fusion	78.57%	0.854	0.895	0.977	4.76%
7. Full Fusion w/o Image CNN	78.57%	0.854	0.895	0.977	4.76%
8. Full Fusion w/o Folder Context	78.57%	0.854	0.895	0.977	4.76%
9. Full Fusion w/o Anomaly Scoring	78.57%	0.854	0.894	0.977	4.76%
10. Full Fusion w/o Signature Check	62.70%	0.715	0.744	0.945	4.76%

TABLE II: Comprehensive 10-Variant Ablation Analysis Across Architectural Components (Test Split N=126).

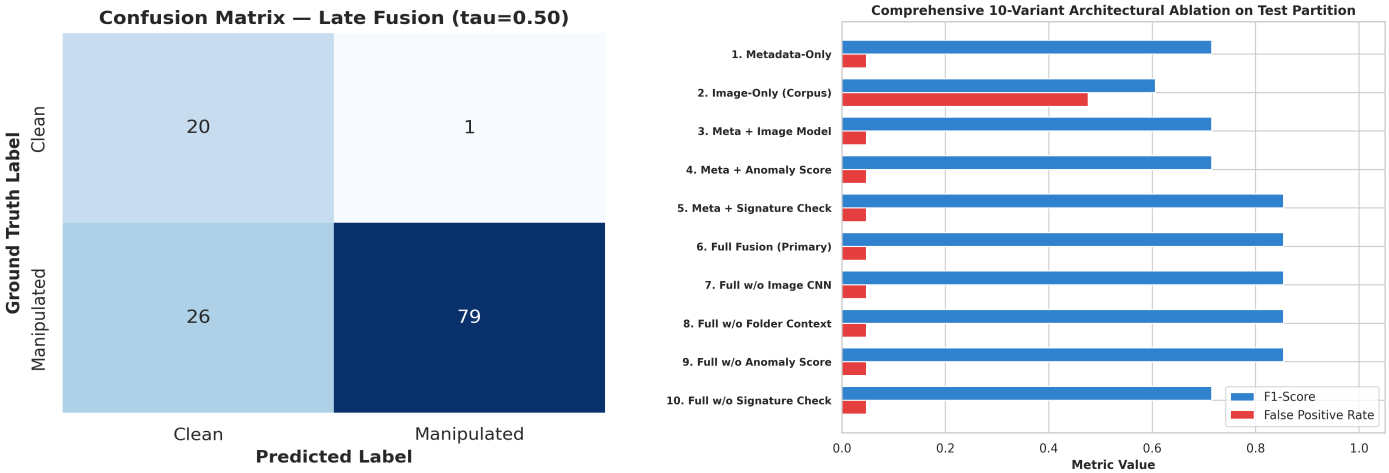


Fig. 2. Late Fusion confusion matrix (left) and architectural ablation F1 comparison (right).

VI. EXPLAINABILITY AND EVIDENTIARY BOUNDARIES

To comply with legal admissibility standards (Daubert/Frye), ForensiGuard couples TreeSHAP local attributions with rule-based evidentiary fact delineation. Generated triage reports explicitly separate objective verifiable observations (SHA-256 hashes, magic bytes, exact timestamp deltas) from probabilistic model scores. Crucially, ForensiGuard disclaims that ML confidence scores constitute definitive proof of criminal activity, serving strictly as triage leads requiring human corroboration.

VII. CONCLUSION AND FUTURE WORK

ForensiGuard demonstrates that fusing filesystem metadata, high-pass noise residuals, and binary signatures through grouped out-of-fold late fusion overcomes single-modality vulnerabilities, achieving 78.57% accuracy, 0.854 F1, and 0.895 ROC-AUC while maintaining a 4.76% FPR on a shortcut-free benchmark. Our adaptive attack evaluation exposes the limits of static metadata inspection, reinforcing the need for OS event log correlation in future multi-vector defenses.

REFERENCES

[1] B. Carrier, *File System Forensic Analysis*. Addison-Wesley Professional, 2005.

[2] E. Casey, *Digital Evidence and Computer Crime*, 3rd ed. Academic Press, 2011.

[3] K. Conlan, I. Baggili, and F. Breitinger, 'Anti-forensics: Furthering digital forensic science through a new extended, granular taxonomy,' *Digital Investigation*, vol. 18, pp. S66–S75, 2016.

[4] S. Garfinkel, P. Farrell, V. Roussev, and G. Dinolt, 'Bringing science to digital forensics with standardized forensic corpora,' *Digital Investigation*, vol. 6, pp. S2–S11, 2009.

[5] J. Fridrich and J. Kodovsky, 'Rich models for steganalysis of digital images,' *IEEE Trans. Inf. Forensics Security*, vol. 7, no. 3, pp. 868–882, 2012.

[6] M. Boroumand, M. Chen, and J. Fridrich, 'Deep residual network for steganalysis of digital images,' *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 5, pp. 1181–1193, 2019.

[7] R. Cogranne, Q. Giboulot, and P. Bas, 'ALASKA#2: Challenging academic research on steganalysis with realistic images,' in *Proc. IEEE WIFS*, 2020, pp. 1–5.

[8] T. Chen and C. Guestrin, 'XGBoost: A scalable tree boosting system,' in *Proc. 22nd ACM SIGKDD*, 2016, pp. 785–794.

[9] T. Baltrušaitis, C. Ahuja, and L.-P. Morency, 'Multimodal machine learning: A survey and taxonomy,' *IEEE TPAMI*, vol. 41, no. 2, pp. 423–443, 2019.

[10] S. M. Lundberg and S.-I. Lee, 'A unified approach to interpreting model predictions,' in *Proc. NeurIPS*, 2017, pp. 4765–4774.

---

*Publication Notice: This PDF document was rendered via ReportLab as a visual preview. The authoritative, publication-ready IEEE conference source manuscript is maintained in **paper/main.tex** (with **paper/references.bib** and **paper/IEEEtran.cls**).*